

From Gaussian RVs to jointly Gaussian vectors

In one dimension, $X \sim \mathcal{N}(\mu, \sigma^2)$ has density

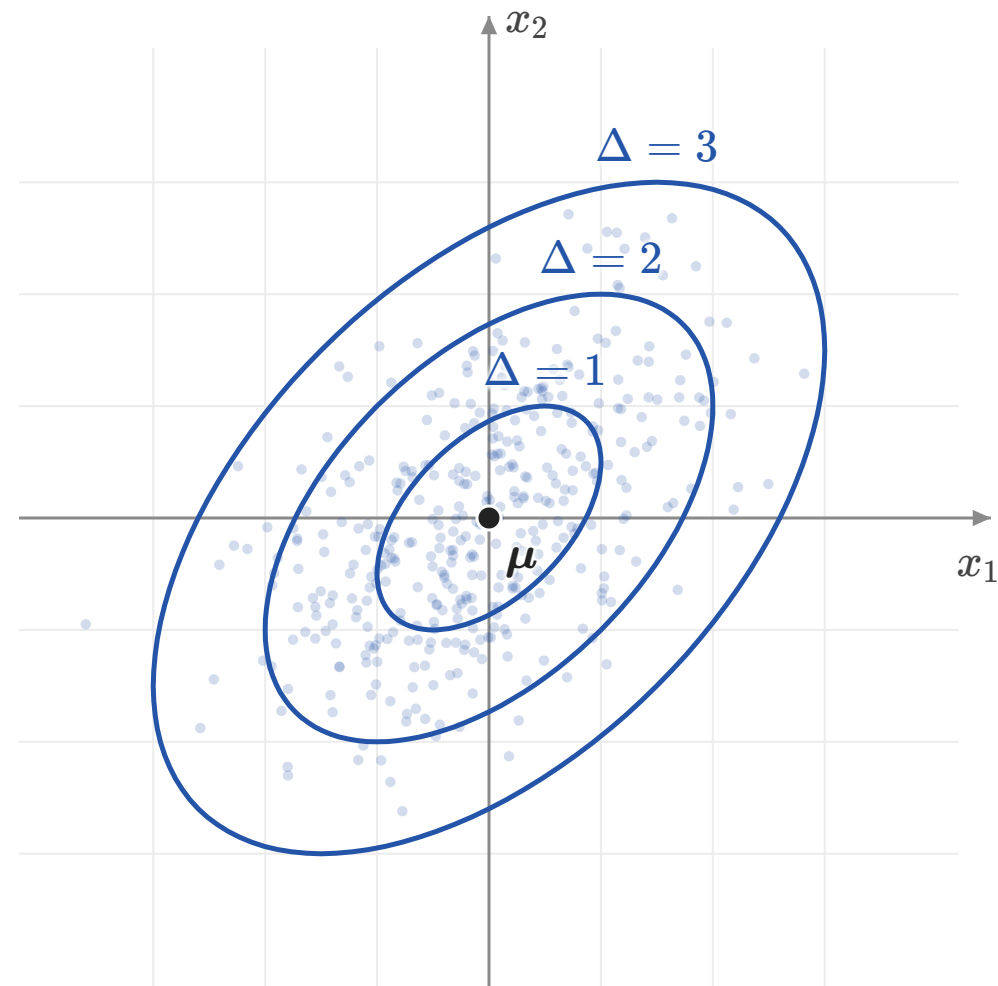
$$p_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-z^2/2}, \quad z = \frac{x - \mu}{\sigma}.$$

Distance from the mean is measured in standard deviations: $|z|$ scales $|x - \mu|$ by the spread σ . The density falls exponentially in z^2 .

In more dimensions the distance is scaled as well, by a different amount in each direction, because the spread differs from one direction to another. This is the Mahalanobis distance:

$$\Delta(\mathbf{x}, \boldsymbol{\mu}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{K}^{-1} (\mathbf{x} - \boldsymbol{\mu})}.$$

Let's see how this distance behaves for different $\mathbf{K}$.



Mahalanobis distance for a diagonal covariance

A diagonal covariance matrix means uncorrelated coordinates, with variances σ_1^2 and σ_2^2 :

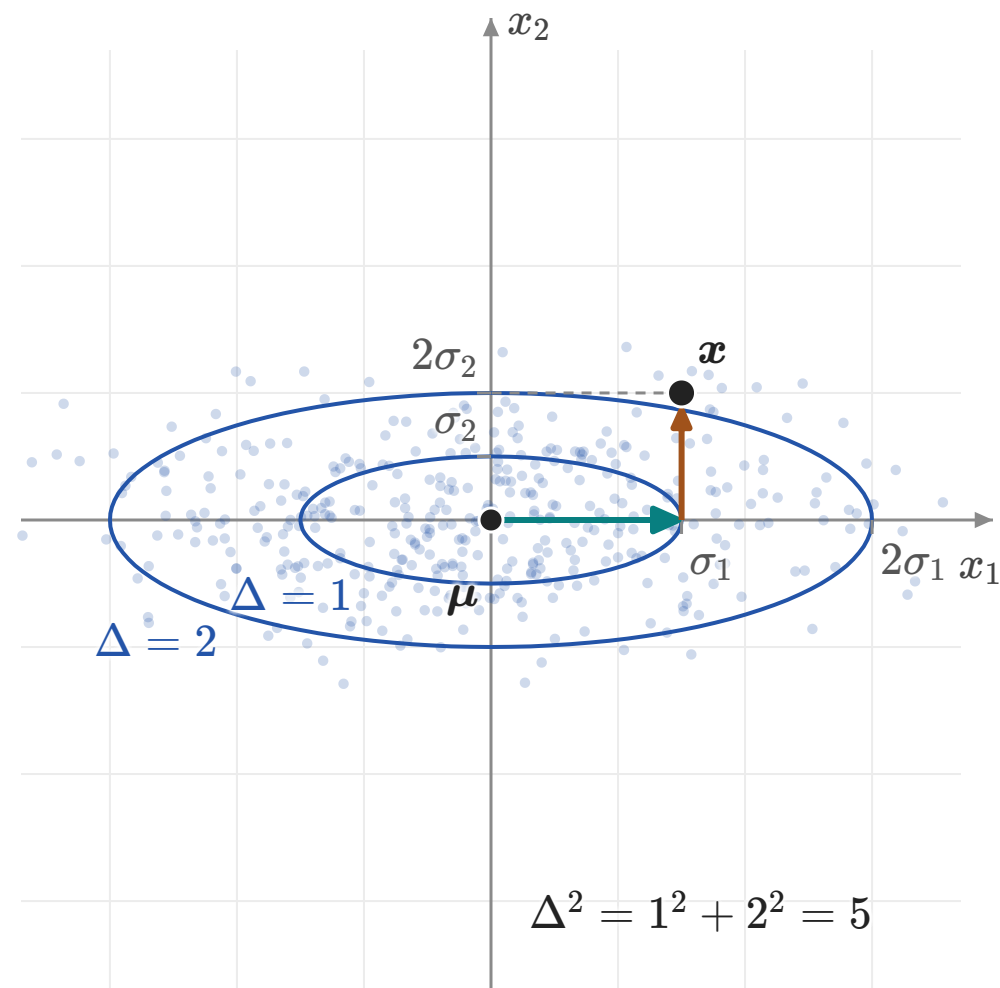
$$\mathbf{K} = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix}, \quad \mathbf{K}^{-1} = \begin{pmatrix} 1/\sigma_1^2 & 0 \\ 0 & 1/\sigma_2^2 \end{pmatrix}.$$

The definition then gives

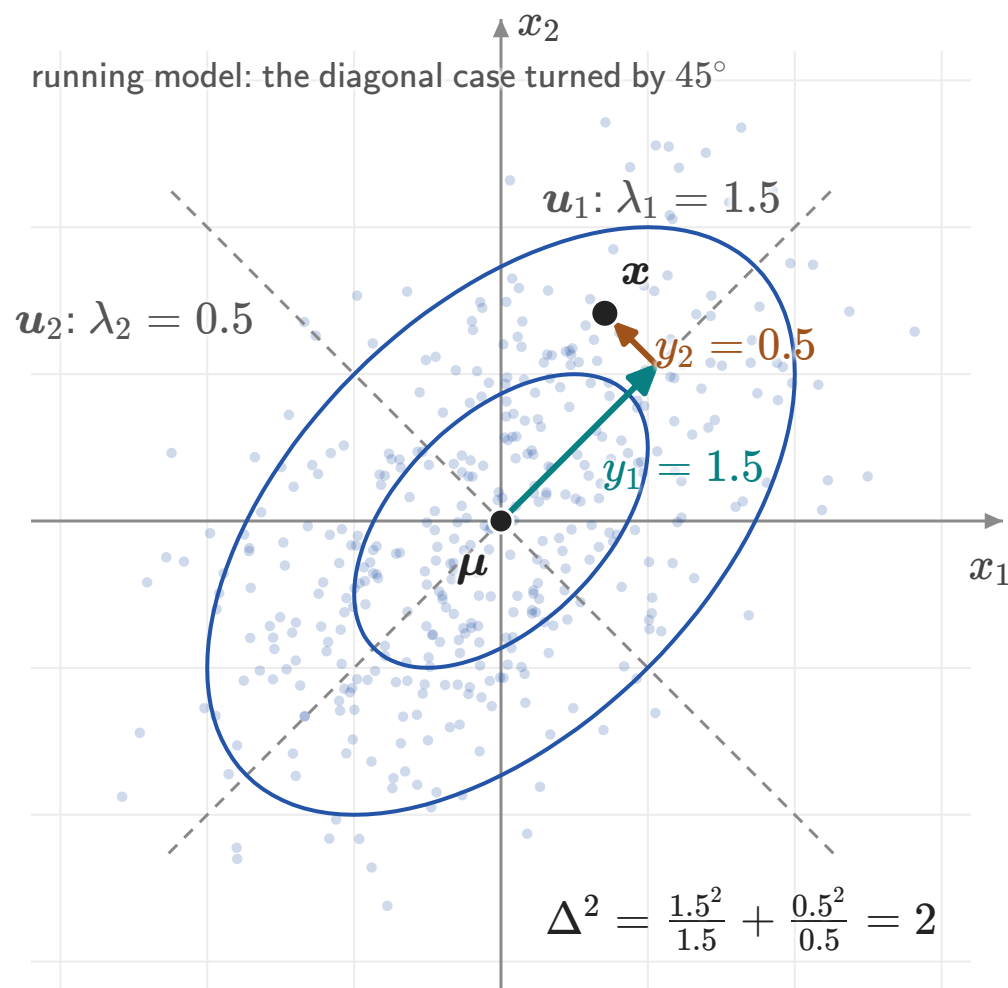
$$\begin{aligned} \Delta^2 &= (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{K}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\ &= \frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2}. \end{aligned}$$

Each coordinate is measured in its own standard deviations

Here $\sigma_1 = 1.5$ and $\sigma_2 = 0.5$. The marked point is 1.5 from the mean along x_1 , one standard deviation, and 1 along x_2 , two standard deviations: $\Delta^2 = 1^2 + 2^2 = 5$.



Mahalanobis distance for a general covariance



Allowing off-diagonal entries in K rotates the axes as well.

K then defines a new set of axes, its principal directions u_1, u_2 (the eigenvectors of K), with variances λ_1, λ_2 along them (its eigenvalues).

The distance is computed in these axes: with y_1, y_2 the coordinates of $x - \mu$ along them, each measured in its own standard deviations,

$$\Delta^2 = \frac{y_1^2}{\lambda_1} + \frac{y_2^2}{\lambda_2}.$$

K defines the axes used to compute the distance

A diagonal K keeps the coordinate axes, and off-diagonal entries rotate them.

The multivariate Gaussian density

$$p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\mathbf{K}|^{1/2}} \exp \left(-\frac{1}{2} \Delta^2(\mathbf{x}, \boldsymbol{\mu}) \right)$$

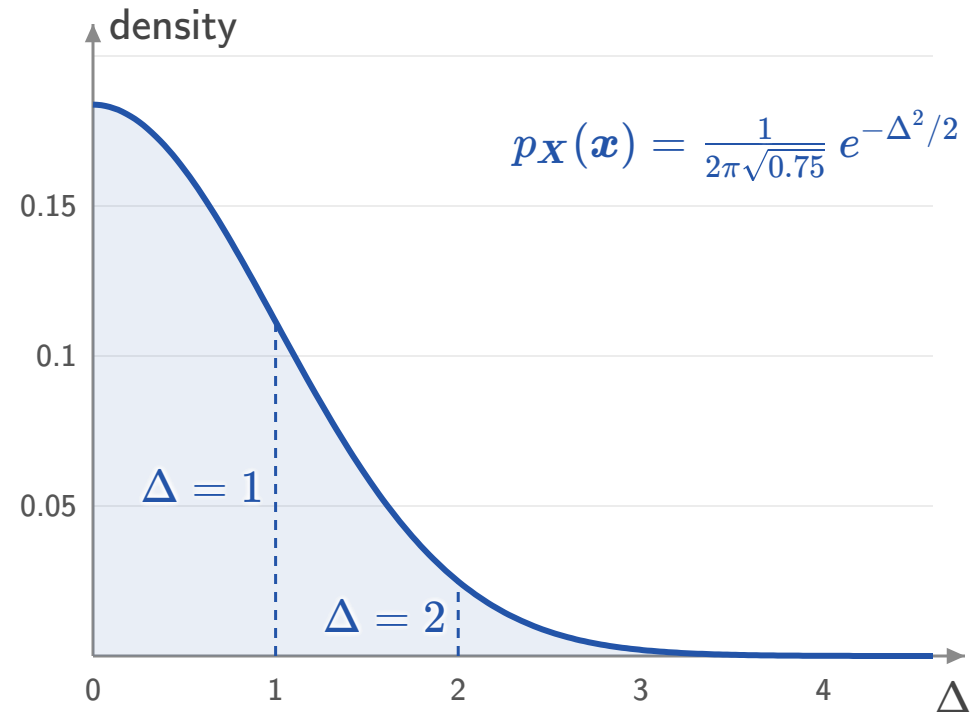
This is the one-dimensional density with z^2 replaced by Δ^2 and σ by $|\mathbf{K}|^{1/2}$, for positive definite $\mathbf{K}$.

Written in terms of $\mathbf{K}$,

$$p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\mathbf{K}|^{1/2}} e^{-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{K}^{-1} (\mathbf{x} - \boldsymbol{\mu})}.$$

Contours of equal density are Mahalanobis ellipses

Within one Gaussian, a larger Mahalanobis distance means a lower density. Right: the running model, with $|\mathbf{K}| = 0.75$.



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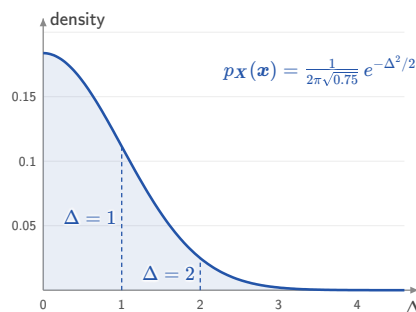
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Within one Gaussian, a larger Mahalanobis distance means a lower density. Right: the running model, with $|\mathbf{K}| = 0.75$.



Derivation

Why $|\mathbf{K}|^{1/2}$ takes the place of σ : rotating to the principal coordinates, $\mathbf{y} = \mathbf{U}^T(\mathbf{x} - \boldsymbol{\mu})$ with $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_d)$ orthonormal, changes no volume and gives $\Delta^2 = \sum_i y_i^2 / \lambda_i$. The determinant is the product of the eigenvalues, $|\mathbf{K}| = \prod_i \lambda_i$, so the density is a product of one-dimensional Gaussian densities with variances λ_i :

$$\frac{1}{(2\pi)^{d/2} |\mathbf{K}|^{1/2}} e^{-\Delta^2/2} = \prod_{i=1}^d \frac{1}{\sqrt{2\pi\lambda_i}} e^{-y_i^2/2\lambda_i}.$$

Running model: $|\mathbf{K}| = 1 \cdot 1 - 0.5^2 = 0.75 = 1.5 \times 0.5$.

Two points at the same Euclidean distance

Running model: $\mu = \mathbf{0}$, $K = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$. In

(x_1, x_2) order, A is $(2, 2)$ and B is $(-2, 2)$, both $\sqrt{8}$ from the mean.

With $K^{-1} = \frac{1}{3} \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$,

$\Delta^2 = \frac{1}{3} (4x_1^2 - 4x_1x_2 + 4x_2^2)$, so

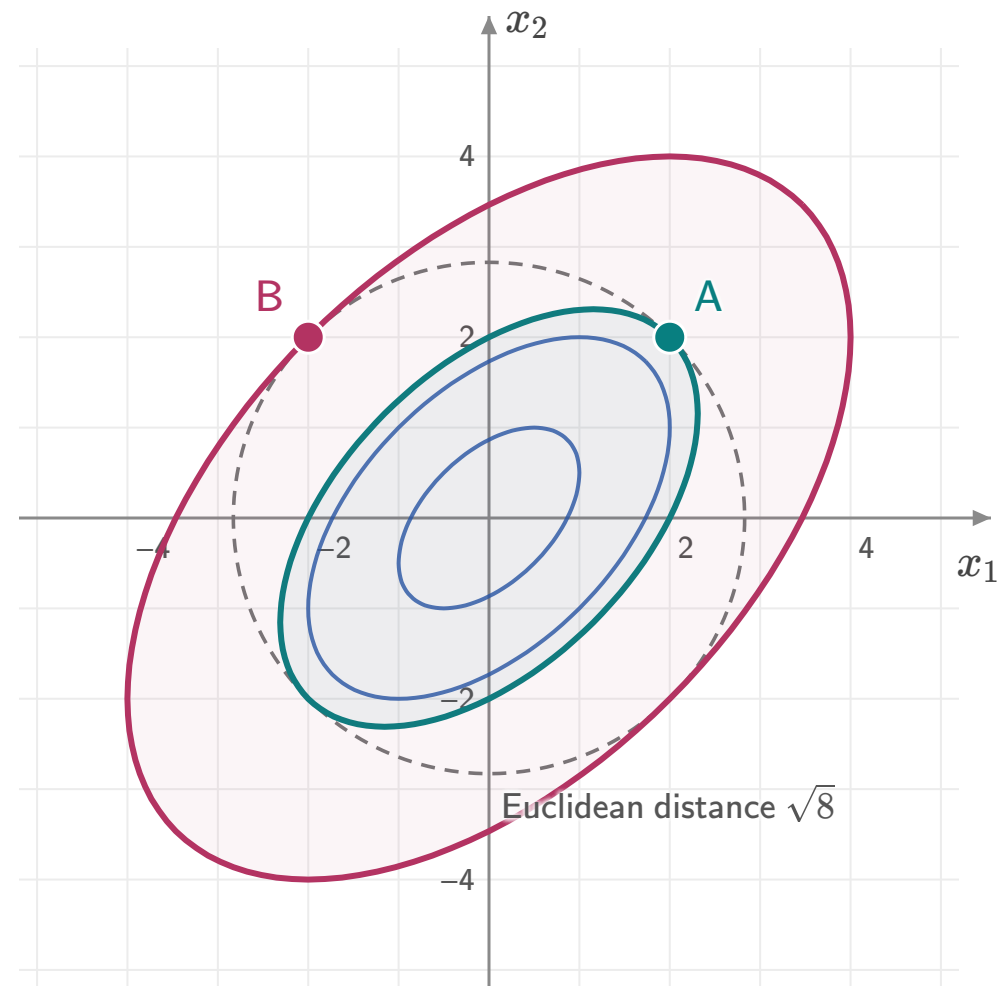
$$\Delta_A^2 = \frac{1}{3} (16 - 16 + 16) = \frac{16}{3},$$

$$\Delta_B^2 = \frac{1}{3} (16 + 16 + 16) = 16.$$

$$p_X(A) \approx 0.0128, \quad p_X(B) \approx 6.2 \times 10^{-5}.$$

Euclidean distance does not measure how likely a point is

The density at A is $e^{(16-16/3)/2} \approx 207$ times that at B.



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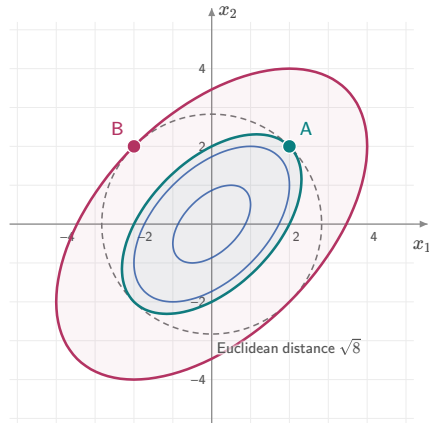
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Derivation

For a 2×2 matrix, $\begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} = \frac{1}{ac-b^2} \begin{pmatrix} c & -b \\ -b & a \end{pmatrix}$. Here

$|K| = 1 - 0.25 = 0.75$, so

$$K^{-1} = \frac{1}{0.75} \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}.$$

From the density with $d = 2$ and $|K|^{1/2} = \sqrt{0.75}$:

$$p_X(A) = \frac{e^{-(16/3)/2}}{2\pi\sqrt{0.75}} \approx 0.0128$$

$$p_X(B) = \frac{e^{-16/2}}{2\pi\sqrt{0.75}} \approx 6.2 \times 10^{-5}$$

$$\frac{p_X(A)}{p_X(B)} = e^{(16-16/3)/2} = e^{16/3} \approx 207$$

Linear combinations of jointly Gaussian RVs are jointly Gaussian

Jointly Gaussian

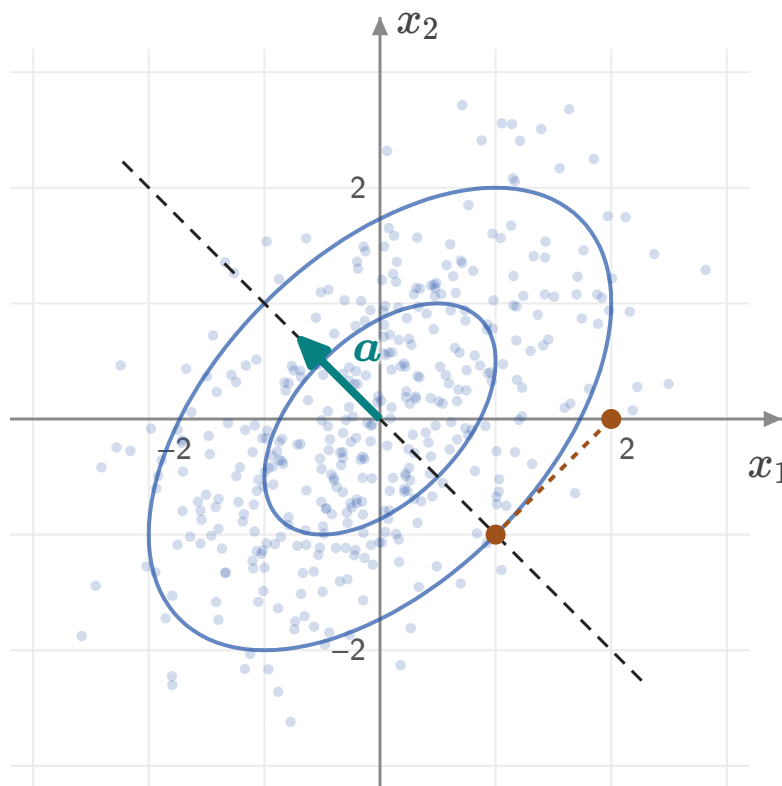
Every linear combination of the coordinates is Gaussian: for every $\mathbf{a}$

$$\mathbf{a}^T \mathbf{X} \sim \mathcal{N}(\mathbf{a}^T \boldsymbol{\mu}, \mathbf{a}^T \mathbf{K} \mathbf{a}).$$

Since its mean and variance can be computed from $\boldsymbol{\mu}$ and $\mathbf{K}$, this gives its distribution.

Predict: in which direction of $\mathbf{a}$ is the bell widest?

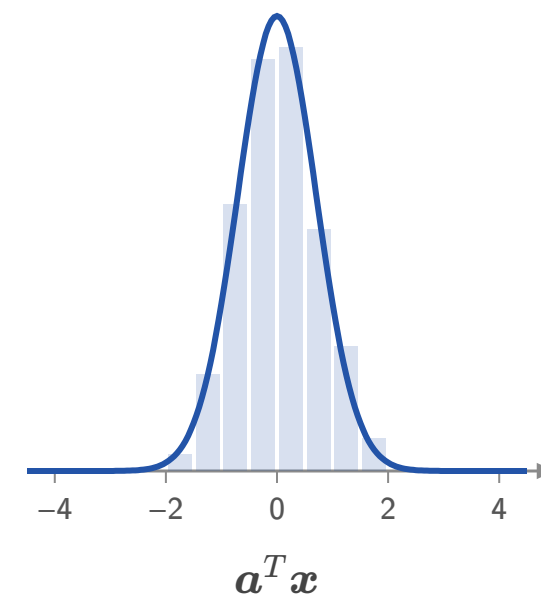
unit $\mathbf{a}$	$\mathbf{a}^T \mathbf{K} \mathbf{a}$
$(1, 0)^T$	1
$(1, 1)^T / \sqrt{2}$	1.5
$(-1, 1)^T / \sqrt{2}$	0.5



density; bars: the 400 dots

$$\mathbf{a} = (-0.71, 0.71)^T$$

$$\mathbf{a}^T \mathbf{K} \mathbf{a} = 0.5$$



Linear combinations of jointly Gaussian RVs are jointly Gaussian

Jointly Gaussian

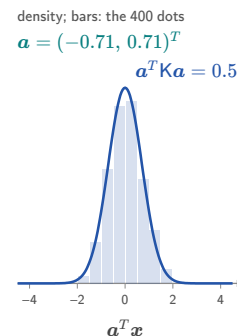
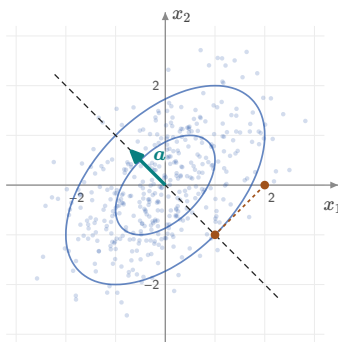
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unit $\mathbf{a}$	$\mathbf{a}^T \mathbf{K} \mathbf{a}$
$(1, 0)^T$	1
$(1, 1)^T / \sqrt{2}$	1.5
$(-1, 1)^T / \sqrt{2}$	0.5



Derivation

$\mathbf{X}$ is jointly Gaussian, so $\mathbf{a}^T \mathbf{X}$ is Gaussian, and its mean and variance determine it.

Put the single row $\mathbf{A} = \mathbf{a}^T$ into $\mathbb{E}[\mathbf{A}\mathbf{X}] = \mathbf{A}\boldsymbol{\mu}$ and $\text{Cov}(\mathbf{A}\mathbf{X}) = \mathbf{A}\mathbf{K}\mathbf{A}^T$: the mean is $\mathbf{a}^T \boldsymbol{\mu}$ and the variance is $\mathbf{a}^T \mathbf{K} \mathbf{a}$.

For $\mathbf{a} = (a_1, a_2)^T$ and the running $\mathbf{K}$,

$$\mathbf{a}^T \mathbf{K} \mathbf{a} = a_1^2 + a_2^2 + 2(0.5) a_1 a_2 = 1 + a_1 a_2$$

for a unit vector, since $a_1^2 + a_2^2 = 1$. Hence

$$(1, 0)^T : 1 + 0 = 1$$

$$(1, 1)^T / \sqrt{2} : 1 + \frac{1}{2} = 1.5$$

$$(-1, 1)^T / \sqrt{2} : 1 - \frac{1}{2} = 0.5$$

$1 + a_1 a_2$ is largest at $a_1 = a_2 = 1/\sqrt{2}$: the bell is widest along $(1, 1)^T$.

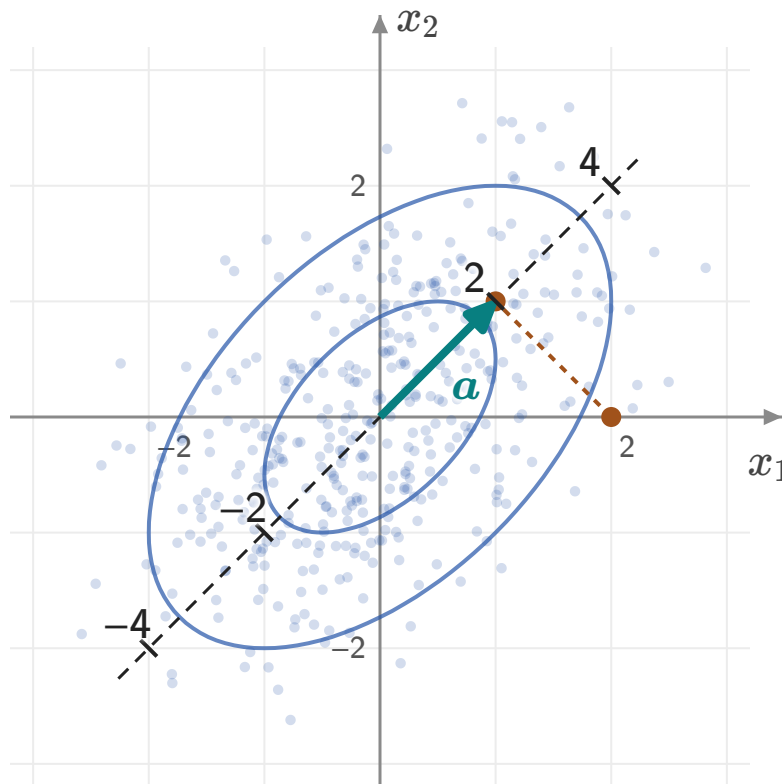
Variance of a sum of coordinates

For the running model,

$$X_1 + X_2 \sim \mathcal{N}(0, \underbrace{1 + 1 + 2(0.5)}_3).$$

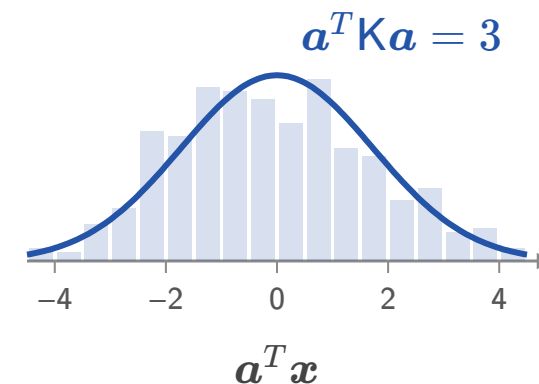
This is $\mathbf{a}^T \mathbf{X}$ with $\mathbf{a} = (1, 1)^T$, of length $\sqrt{2}$: the variance is $2 \times 1.5 = 3$.

The arrow is $\mathbf{a}$, and the ticks on the dashed line mark $\mathbf{a}^T \mathbf{x}$. A longer $\mathbf{a}$ packs the ticks closer, so the same cloud projects to a wider spread.



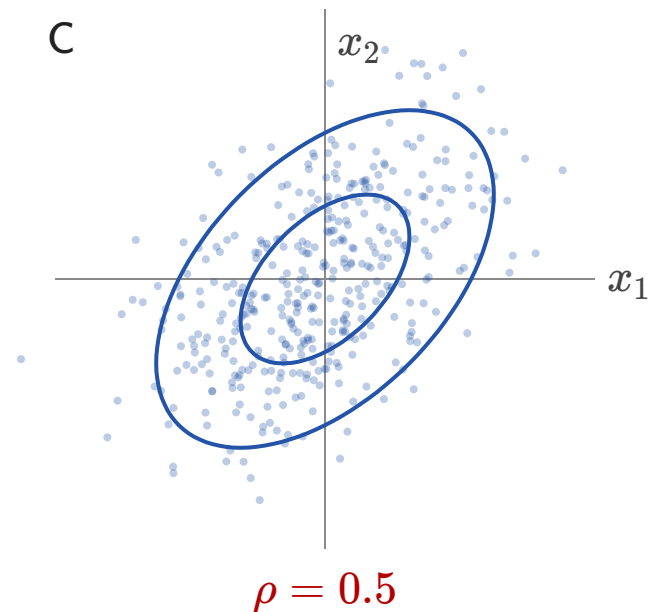
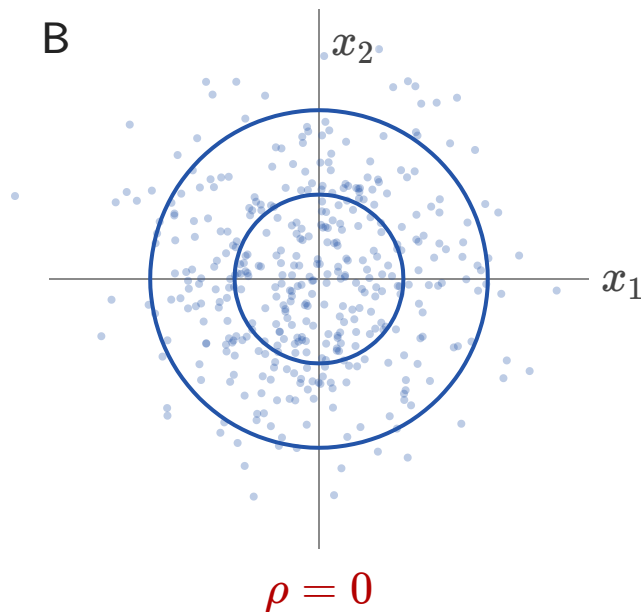
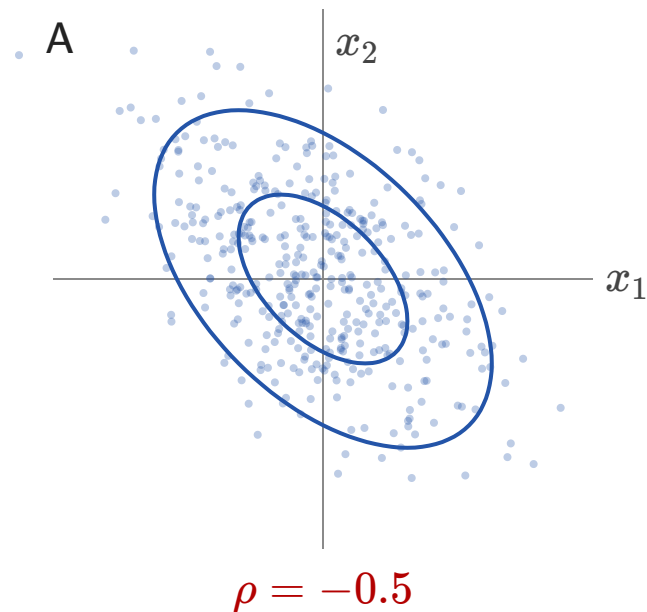
density; bars: the 400 dots

$$\mathbf{a} = (1, 1)^T$$



Correlation and the shape of the cloud

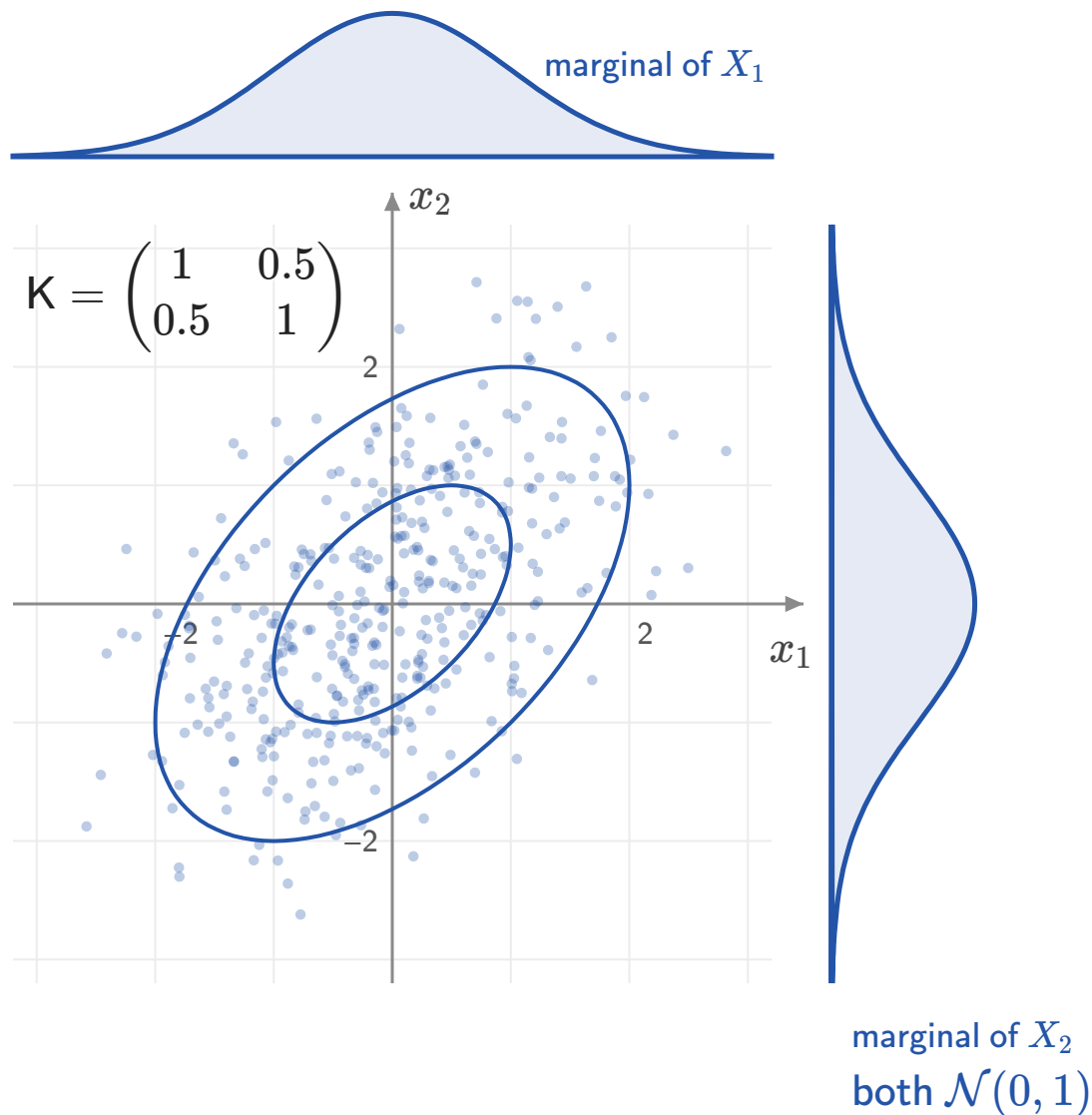
Question: Match each cloud to $\rho = -0.5$, 0, or 0.5.



All three models have $X_1 \sim \mathcal{N}(0, 1)$ and $X_2 \sim \mathcal{N}(0, 1)$, with $\mathbf{K} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$.

The covariance determines the tilt. The marginal variances stay at 1.

Marginals of a bivariate Gaussian



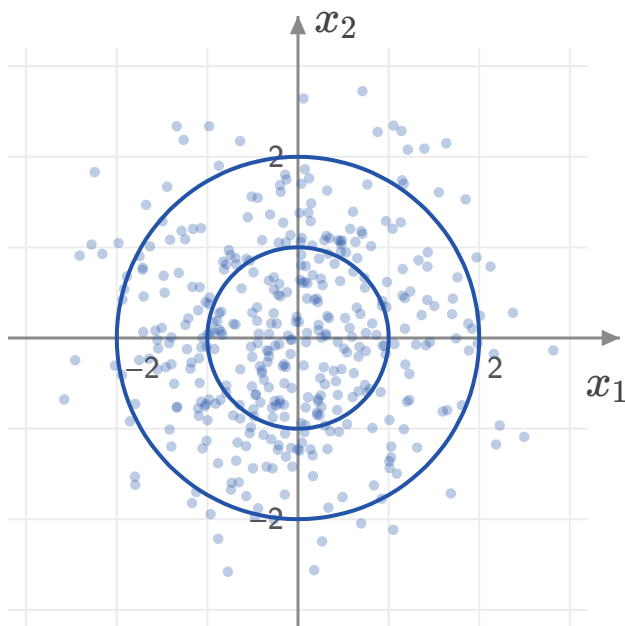
Each marginal is Gaussian: $X_i = \mathbf{e}_i^T \mathbf{X}$ is a linear combination.

ρ	tilt	$\text{Var}(X_1)$	$\text{Var}(X_2)$
-0.5	down	1	1
0	none	1	1
0.5	up	1	1

Correlation tilts the cloud and leaves the marginals fixed

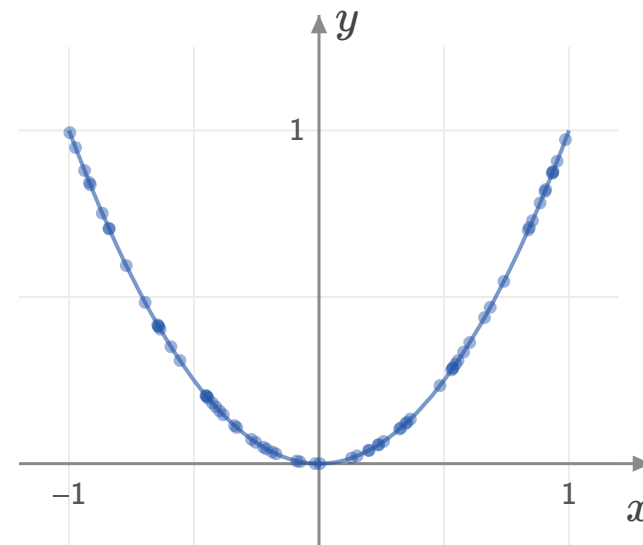
Each marginal alone cannot tell these models apart. The relationship lives in the joint distribution, through the off-diagonal of K . Gaussian marginals alone do not make a vector jointly Gaussian.

Zero covariance and independence



Jointly Gaussian, $\rho = 0$

$\mathbf{K} = \mathbf{I}$, so $\text{Cov}(X_1, X_2) = 0$.



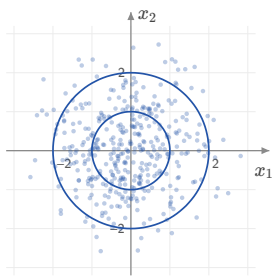
A curved relationship

$X \sim \text{Uni}(-1, 1)$ and $Y = X^2$:

$$\text{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X] \mathbb{E}[X^2] = 0.$$

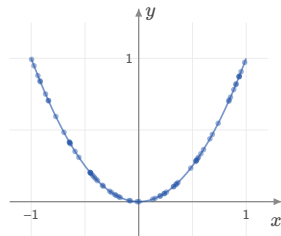
Question: In which picture does observing one coordinate help predict the other?

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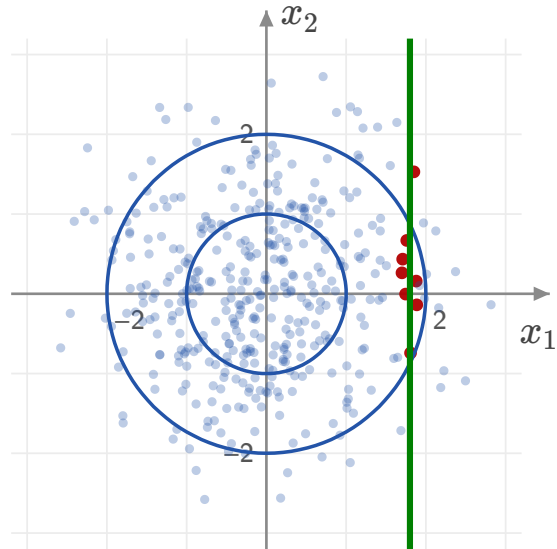
Derivation

The density of $X \sim \text{Uni}(-1, 1)$ is symmetric about 0, so its odd moments vanish: $\mathbb{E}[X] = \int_{-1}^1 \frac{1}{2}x \, dx = 0$ and $\mathbb{E}[X^3] = \int_{-1}^1 \frac{1}{2}x^3 \, dx = 0$. With $Y = X^2$,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] = \mathbb{E}[X^3] - \mathbb{E}[X] \mathbb{E}[X^2] = 0.$$

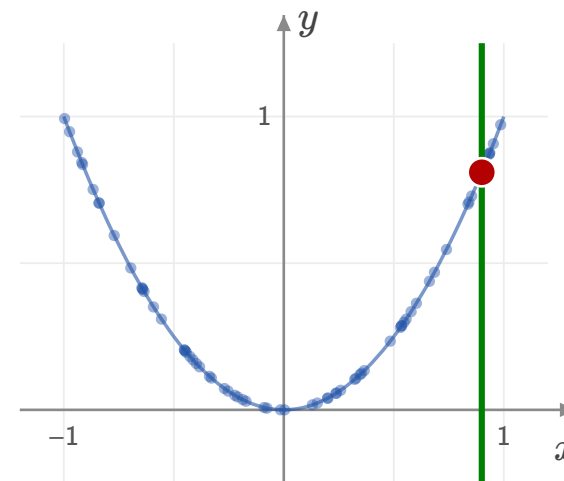
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Uncorrelated versus independent



$$X_2 | X_1 = 1.8 \\ \sim \mathcal{N}(0, 1)$$

the same for every x_1



$$Y | X = 0.9 \\ \text{equals } 0.81$$

exactly, with no spread

Jointly Gaussian, $\rho = 0$

A diagonal covariance matrix implies independent components. Knowing one coordinate leaves the other's distribution unchanged.

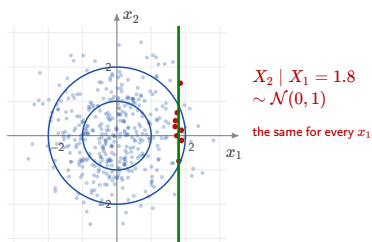
A curved relationship

$\text{Cov}(X, Y) = 0$, yet knowing X determines Y exactly.

Zero covariance does not imply independence, except for jointly Gaussian RVs

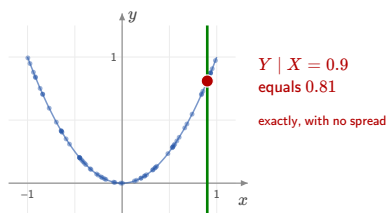
Independence always implies zero covariance. The converse holds for jointly Gaussian RVs, whose density factors when K is diagonal (left), and fails in general (right).

Uncorrelated versus independent



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A diagonal covariance matrix implies independent components. Knowing one coordinate leaves the other's distribution unchanged.



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Zero covariance does not imply independence, except for jointly Gaussian RVs

Independence always implies zero covariance. The converse holds for jointly Gaussian RVs, whose density factors when $\mathbf{K}$ is diagonal (left), and fails in general (right).

Derivation

Jointly Gaussian with $\rho = 0$ means $\mathbf{K} = \text{diag}(\sigma_1^2, \sigma_2^2)$. Then $\mathbf{K}^{-1} = \text{diag}(1/\sigma_1^2, 1/\sigma_2^2)$ and $|\mathbf{K}| = \sigma_1^2 \sigma_2^2$, so the exponent is a sum and the density a product:

$$\begin{aligned} p_{\mathbf{X}}(\mathbf{x}) &= \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(x_1 - \mu_1)^2}{2\sigma_1^2} - \frac{(x_2 - \mu_2)^2}{2\sigma_2^2}\right) \\ &= \frac{e^{-(x_1 - \mu_1)^2/2\sigma_1^2}}{\sqrt{2\pi}\sigma_1} \cdot \frac{e^{-(x_2 - \mu_2)^2/2\sigma_2^2}}{\sqrt{2\pi}\sigma_2} = p_{X_1}(x_1) p_{X_2}(x_2). \end{aligned}$$

The joint density is the product of the marginals, so X_1 and X_2 are independent. Here $\mathbf{K} = \mathbf{I}$: $\sigma_1 = \sigma_2 = 1$.